

A+ Core 2 Memorization Sheet

Print this and keep it beside you. Core 2 is about names: which Windows tool, which command, which edition, which security term, which step comes next, which symptom points where. Every line here is fair game on the exam.

01 The exam itself

FACT	VALUE
Exam	CompTIA A+ Core 2, 220-1202 (A+ needs Core 1 220-1201 and Core 2)
Questions and time	Up to 90 questions in 90 minutes; multiple choice plus performance-based items
Passing	700 on a scale of 100 to 900 (Core 1 needs 675)
Operating Systems	28%
Security	28%
Software Troubleshooting	23%
Operational Procedures	21%

Operating systems and security are more than half the exam. Know every tool by its file name, every command by its job, and every procedure in its exact order.

02 Operating systems, filesystems, and Windows editions

OPERATING SYSTEM	FACTS
Windows	Workstation OS ; Home, Pro, Pro for Workstations, Enterprise; NTFS ; domain or workgroup; the only one with Group Policy and BitLocker
Linux	Open-source kernel with distributions (Ubuntu and Debian use apt, Fedora and RHEL use dnf); ext4 or XFS ; case-sensitive; root account; systemd
macOS	Apple desktops and laptops only; APFS ; App Store; Time Machine; FileVault; Keychain; Spotlight
Chrome OS	Google's browser-centered OS on Chromebooks; cloud storage, Android apps, managed from the Google Admin console
iOS and iPadOS	Apple phones and tablets; closed app store; no sideloading without enterprise profiles
Android	Google's open mobile OS ; many vendors; sideloading possible; more fragmentation of updates
Vendor life cycle	End-of-life (EOL): no more security updates; update limitations: hardware or edition blocks a version; Windows 10 support ended October 2025
Compatibility	Applications and drivers are built per OS and per architecture; 32-bit apps run on 64-bit Windows, not the reverse; files move between OSes only on filesystems both can read (exFAT , FAT32)

FILESYSTEM	WHERE	REMEMBER
NTFS	Windows system and data drives	Permissions, encryption (EFS), compression, journaling, huge volumes and files
ReFS	Windows Server and Pro for Workstations	Resilient: integrity streams, self-healing, very large volumes; not bootable in most editions
FAT32	Old drives, small flash, firmware	Universal compatibility; 4 GB maximum file size; Windows formats only up to 32 GB

FILESYSTEM	WHERE	REMEMBER
exFAT	Large flash drives and SD cards	Big files, no 4 GB limit, readable by Windows, macOS , and Linux; no permissions or journaling
ext4	Linux default	Journaling, large files; Windows cannot read it without extra software
XFS	Linux (RHEL default), servers	High performance with very large files and volumes
APFS	macOS and iOS	Snapshots, encryption, cloning, SSD optimized; replaced HFS+

WINDOWS EDITION	WHAT IT ADDS
Home (10 and 11)	Consumer; workgroup only, no domain join; no BitLocker, no Remote Desktop host, no Group Policy Editor (gpedit.msc); 128 GB RAM limit
Pro (10 and 11)	Domain join, BitLocker, Remote Desktop host, gpedit.msc, Hyper-V; 2 TB RAM
Pro for Workstations (10)	ReFS, persistent memory, up to four CPUs , 6 TB RAM ; for high-end workstations
Enterprise (10 and 11)	Volume licensing; adds AppLocker, DirectAccess, advanced management; 6 TB RAM
N versions	European editions without Media Player and media features
Desktop and RDP	Windows 11 has the centered taskbar and new Settings; every edition can be an RDP client, only Pro and above can be an RDP host
Upgrade paths	In-place upgrade keeps files, apps, and settings (Windows 10 to 11 if the hardware qualifies; Home to Pro with a key); clean install wipes; 32-bit to 64-bit requires a clean install
Windows 11 hardware	TPM 2.0 , UEFI with Secure Boot capable, 64-bit dual-core 1 GHz CPU , 4 GB RAM , 64 GB storage

03 Installation, upgrades, and application requirements

BOOT OR INSTALL ITEM	FACTS
Boot methods	USB , network (PXE , Preboot Execution Environment), solid-state or flash drive, internet-based (cloud recovery), external or hot-swappable drive, internal drive partition, multiboot (boot menu chooses the OS)
Clean install	Wipes the drive; new OS ; restore files afterward
Upgrade (in-place)	Keeps files, apps, and settings; needs a supported path and compatible hardware and drivers
Image deployment	A prepared disk image copied to many machines (cloning, MDT , Configuration Manager)
Remote network installation	Boot from PXE , install from a server share
Zero-touch deployment	Device ships to the user and enrolls itself through the vendor's service (Windows Autopilot); no technician touches it
Recovery partition	Vendor partition that restores the factory image
Repair installation	Reinstalls Windows over itself, keeping data and apps, to fix system files
Third-party drivers	Load storage or RAID drivers during setup when the installer cannot see the disk
GPT	GUID partition table: UEFI , up to 128 partitions, disks over 2 TB
MBR	Master boot record: legacy BIOS , four primary partitions (or three plus an extended), 2 TB limit

BOOT OR INSTALL ITEM	FACTS
Upgrade considerations	Back up files and user preferences; check application and driver support and backward compatibility; hardware compatibility; feature updates twice a year become once a year; product life cycle (EOL date)

Application requirements and impact

- 32-bit versus 64-bit: a 64-bit app needs a 64-bit **OS**; a 32-bit app runs on either; 32-bit is limited to about 4 **GB** of addressable memory
- Dedicated graphics card with its own **VRAM** for gaming, **CAD**, and video; integrated graphics shares system **RAM**; check **VRAM**, **RAM**, **CPU**, and storage minimums against the machine, not just the **OS**
- External hardware tokens: some software licenses require a **USB** dongle or key present to run
- Distribution: physical media, a mountable **ISO** file, a downloadable package, or image deployment; verify the download's hash
- Impact considerations: device (resources, compatibility), network (bandwidth, ports, licensing servers), operation (training, downtime, workflows), business (cost, compliance, support)
- Cloud productivity: email systems, storage with sync and folder settings, collaboration tools (spreadsheets, videoconferencing, presentations, word processing, instant messaging), identity synchronization with the directory, licensing assignment per user

04 Windows tools by file name

TOOL	OPEN WITH	USE IT FOR
Task Manager	Ctrl+Shift+Esc; taskmgr	Processes (end a hung app, see CPU , memory, disk, network), Performance graphs, Startup apps, Users, Services
Event Viewer	eventvwr.msc	Application, Security, System logs; errors and warnings behind crashes and failed services
Disk Management	diskmgmt.msc	Initialize, partition, format, extend, shrink, assign letters, convert MBR and GPT , dynamic disks
Task Scheduler	taskschd.msc	Run scripts and programs at a time or on an event
Device Manager	devmgmt.msc	Drivers: update, roll back, disable; yellow triangle means a problem; resource conflicts
Certificate Manager	certmgr.msc	View, import, and remove user certificates; trusted roots
Local Users and Groups	lusrmgr.msc	Create accounts, reset passwords, group membership (Pro and above)
Performance Monitor	perfmon.msc	Counters over time; data collector sets; baselines
Group Policy Editor	gpedit.msc	Local policies for the computer and users (Pro and above)
System Information	msinfo32.exe	Hardware, drivers, software inventory; BIOS mode; export a report
Resource Monitor	resmon.exe	Live CPU , memory, disk, network per process; find what holds a file
System Configuration	msconfig.exe	Boot options (safe boot), services, startup (now in Task Manager), tools
Disk Cleanup	cleanmgr.exe	Remove temporary files, old updates, previous installations
Disk Defragmenter	dfrgui.exe	Optimize drives: defragment HDDs , TRIM SSDs ; scheduled weekly
Registry Editor	regedit.exe	Edit the registry hives (HKLM , HKCU); export a backup before changing anything
Microsoft Management Console	mmc.exe	Host for snap-ins; build a custom console

05 Command-line tools

WINDOWS COMMAND	JOB
cd, dir	Change directory; list files (dir /a shows hidden)
md, rmdir	Make a directory; remove a directory (rmdir /s removes contents)
robocopy	Robust copy of folder trees with retries, mirroring (/mir), and permissions; xcopy is the old one
ipconfig	Address, mask, gateway; /all adds MAC and DNS ; /release and /renew for DHCP ; /flushdns clears the resolver cache
ping	Reachability and round trip; -t continuous; ping 127.0.0.1 tests the stack
tracert	Every hop to a destination
pathping	Tracert plus per-hop packet loss statistics over time
netstat	Open connections and listening ports; -a all, -n numeric, -b program names
nslookup	Query DNS for a name or address
net use	Map a drive: net use X: \\server\share; net use * /delete removes mappings
net user	List or manage local accounts; net user name password /add
chkdsk	Check the disk; /f fixes errors; /r locates bad sectors and recovers data
format	Format a volume with a filesystem; destroys data
diskpart	Command-line partitioning: list disk, select disk, clean, create partition, convert gpt
hostname, whoami, winver	Computer name; current user and groups; Windows version dialog
[command] /?	Help for any command
gpupdate	Apply Group Policy now; /force reapplies everything
gpresult	Show applied policies; /r summary
sfc	System File Checker; sfc /scannow repairs protected system files (DISM /online /cleanup-image /restorehealth repairs the source)
shutdown	shutdown /r /t 0 restarts now; /s shuts down
LINUX COMMAND	JOB
ls, pwd	List files (ls -la shows all with permissions); print the working directory
cd, mv, cp, rm	Change directory; move or rename; copy (cp -r for folders); remove (rm -rf removes a tree; no undo)
chmod, chown	Change permissions (r 4, w 2, x 1; owner group other, so 755 and 644); change owner and group
grep, find	Search text inside files; find files by name, size, or age
cat, man	Print a file; read the manual page for a command
top, ps	Live process view; list processes (ps aux)
df, du	Free space per filesystem; space used by a directory (du -sh)
fsck, mount	Check a filesystem (unmounted); attach a filesystem to a directory (umount detaches)
su, sudo	Switch user (to root with a password); run one command as root (sudo, logged, preferred)
apt, dnf	Package managers: apt on Debian and Ubuntu (apt update, apt install); dnf on Fedora and RHEL
ip, ping, curl, dig, traceroute	ip addr and ip route for addressing; ping; curl fetches URLs ; dig queries DNS ; traceroute lists hops
nano	Simple terminal text editor (vi and vim are the others)

LINUX COMMAND	JOB
<code>/etc/passwd, /etc/shadow</code>	User accounts; hashed passwords (root only)
<code>/etc/hosts, /etc/fstab, /etc/resolv.conf</code>	Static name mappings; filesystems to mount at boot; DNS servers
<code>systemd, kernel, bootloader</code>	Service manager (<code>systemctl start, enable, status</code>); the kernel is the core; GRUB is the bootloader
<code>root</code>	The all-powerful account (UID 0); log in as a normal user and use <code>sudo</code>

06 Control Panel, Settings, and Windows networking

CONTROL PANEL APPLET	WHAT LIVES THERE
Internet Options	Browser security zones, proxy (Connections tab, LAN settings), certificates, home page for legacy Internet Explorer settings
Devices and Printers	Add, remove, set default, printer properties and queues
Programs and Features	Uninstall or change programs; turn Windows features on or off; view installed updates
Network and Sharing Center	Adapter settings, network profile, sharing options, troubleshooting
System	Computer name, domain or workgroup, remote settings, System Protection (restore points), advanced (performance, virtual memory, environment variables)
Windows Defender Firewall	Turn on or off per profile; allow an app; advanced inbound and outbound rules by port
Mail	Outlook profiles and data files
Sound	Playback and recording devices, default device, levels
User Accounts	Change account type, passwords, UAC level, credential manager
Device Manager	The same <code>devmgmt.msc</code>
Indexing Options	What Windows Search indexes and where
Administrative Tools	Shortcuts to the MMC consoles
File Explorer Options	General (open in This PC), View (show hidden files, show file extensions, protected OS files), Search
Power Options	Power plans (Balanced, Power saver, High performance); sleep and hibernate timers; what closing the lid does; fast startup; USB selective suspend; standby; hibernate writes RAM to disk, sleep keeps RAM powered
SETTINGS CATEGORY	WHAT LIVES THERE
System	Display, sound, notifications, power and sleep, storage, about
Devices	Bluetooth, printers, mouse, typing, pen, AutoPlay, USB
Network and Internet	Status, Wi-Fi , Ethernet, VPN , proxy, data usage, metered connections
Personalization	Background, colors, lock screen, themes, start, taskbar
Apps	Apps and features (uninstall), default apps, optional features, startup
Accounts	Your info, email accounts, sign-in options (PIN , Windows Hello), access work or school, family
Time and Language	Date and time, region, language, speech
Gaming	Game bar, captures, game mode

SETTINGS CATEGORY	WHAT LIVES THERE
Ease of Access	Accessibility: narrator, magnifier, high contrast, captions
Privacy	Location, camera, microphone, diagnostics, app permissions
Update and Security	Windows Update, delivery optimization, Windows Security, backup, troubleshoot, recovery, activation, Find My Device

Windows networking

- Domain joined: central accounts and Group Policy from Active Directory; workgroup: local accounts on each PC, small networks
- Shared resources: printers, file servers, mapped drives (net use or Map network drive); UNC paths \\server\share in File Explorer
- Local firewall: allow an application through, add port exceptions, separate rules per profile; public network profile blocks discovery and sharing, private allows it
- Client configuration: IP address, subnet mask, gateway, DNS servers; static (typed) or dynamic (DHCP); alternate configuration
- Connections: VPN (built-in client, add a VPN connection), wireless (SSID and security type), wired, WWAN cellular (SIM, APN, data plan)
- Proxy settings: automatic detection, setup script, or manual address and port; metered connections limit updates and background data

07 macOS and Linux features

MACOS ITEM	FACTS
File types	.dmg is a disk image you mount and drag the app from; .pkg is an installer package; .app is the application bundle itself; App Store installs and updates; uninstall by dragging to Trash or running the vendor uninstaller
System folders	/Applications (apps), /Users (home folders), /Library (system-wide support files), /System (the OS, protected), /Users/name/Library (per-user settings, hidden)
Apple ID and corporate restrictions	Personal Apple ID for iCloud and the App Store; managed Apple IDs and MDM restrict what corporate Macs can do
Best practices	Backups with Time Machine, antivirus, updates and patches, Rapid Security Response (small security patches between full updates)
System Preferences / Settings	Displays, Network, Printers and Scanners, Privacy and Security, Accessibility, Time Machine
Features	Multiple desktops (Spaces) and Mission Control, Keychain (passwords), Spotlight (search), iCloud, iMessage, FaceTime, iCloud Drive, trackpad gestures, Finder (file manager), Dock, Continuity and Handoff between Apple devices
Utilities	Disk Utility (partition, format, First Aid), FileVault (full-disk encryption), Terminal (shell), Force Quit (Command-Option-Esc)

08 Security concepts

Physical security and physical access

- Bollards stop vehicles; access control vestibule (mantrap) admits one person at a time; badge reader; video surveillance; alarm systems; motion sensors; door locks; equipment locks (cable locks); security guards; fences; lighting; magnetometers (metal detectors)
- Physical access: key fobs, smart cards, mobile digital keys, metal keys; biometrics: retina, fingerprint, palm print, facial recognition, voice recognition

LOGICAL SECURITY TERM	MEANING
Principle of least privilege	Every account gets only the rights its job needs
Zero Trust	Never trust by location; verify every user and device for every request
Access control list	The list of who may do what to a resource (files, firewalls, routers)
MFA methods	Email code, hardware token, authenticator application, SMS , voice call, TOTP (time-based one-time password), OTP ; SMS and voice are the weakest
SAML	Security Assertion Markup Language: passes authentication between an identity provider and a web application; the basis of web SSO
SSO	One login for many systems
Just-in-time access	Elevated rights granted only for the task and the time needed
PAM	Privileged access management: vaulted admin credentials, checked out, logged
MDM	Mobile device management: enroll, configure, enforce, wipe
DLP	Data loss prevention: stops sensitive data leaving by email, upload, or USB
IAM	Identity and access management: the whole system of accounts, roles, and authentication
Directory services	Active Directory and LDAP : the database of users, computers, and groups
WIRELESS SECURITY	FACTS
WPA2	AES-CCMP encryption; personal (pre-shared key) or enterprise (802.1X with RADIUS); the minimum acceptable today
WPA3	Stronger: SAE handshake resists offline password guessing, forward secrecy, protected management frames; enterprise 192-bit mode
TKIP	The old WPA encryption; deprecated, weak; never choose it
AES	Advanced Encryption Standard: the strong cipher in WPA2 and WPA3
RADIUS	Central authentication for Wi-Fi and VPN ; UDP 1812 and 1813; encrypts only the password
TACACS+	Cisco device administration authentication; TCP 49; encrypts the whole session; separates authentication, authorization, accounting
Kerberos	Active Directory's ticket-based authentication; the KDC issues tickets; clocks must agree within about five minutes
Multifactor	Something you know, have, or are; two of them
MALWARE	SIGNATURE
Trojan	Pretends to be useful software; installs a backdoor
Rootkit	Hides deep in the OS or firmware; invisible to normal tools; often needs a reinstall
Virus	Attaches to files; spreads when they run
Spyware	Watches and reports activity
Ransomware	Encrypts files and demands payment
Keylogger	Records keystrokes for passwords
Boot sector virus	Infects the boot record; loads before the OS
Cryptominer	Steals CPU and GPU to mine currency; slow and hot machine

MALWARE	SIGNATURE
Stalkerware	Secretly tracks a person's location and messages
Fileless	Lives in memory and legitimate tools (PowerShell); no file for antivirus to scan
Adware and PUP	Ads and unwanted bundled programs; nuisance more than threat
Tools	Recovery console (WinRE), EDR , MDR (managed by a provider), XDR (across endpoints and network), antivirus, antimalware, email security gateway, software firewalls, user education and antiphishing training, OS reinstallation as the last resort
SOCIAL ENGINEERING AND THREATS	SIGNATURE
Phishing, vishing, smishing	Fake email, voice call, SMS ; QR code phishing hides the link in a code
Spear phishing and whaling	Targeted at a person; targeted at an executive
Shoulder surfing, tailgating	Watching a screen or keyboard; following someone through a door
Impersonation, dumpster diving	Pretending to be someone; digging paper out of the trash
DoS and DDoS	Flooding a service; distributed from many bots
Evil twin	A rogue access point with the real network's name
Zero-day	Exploits a flaw with no patch yet
Spoofing, on-path	Faking an address or identity; intercepting between two parties
Brute force, dictionary	Trying every password; trying likely words
Insider threat	An authorized person misusing access
SQL injection, XSS	Database commands through a form; scripts injected into a web page
BEC, supply chain	Business email compromise (fake invoice from the boss); attack through a vendor or software pipeline
Vulnerabilities	Non-compliant, unpatched, unprotected (no antivirus or firewall), EOL systems, BYOD devices

09 Windows security, Active Directory, hardening, and mobile security

WINDOWS SECURITY SETTING	FACTS
Defender Antivirus	Built in; turn on, keep definitions updated; real-time protection
Firewall	On per profile; port rules and application rules; block inbound by default
Accounts	Local versus Microsoft account; Standard user; Administrator; Guest (disabled); Power Users (legacy group)
Sign-in options	Username and password, PIN , fingerprint, facial recognition, SSO , passwordless Windows Hello
NTFS versus share permissions	Share permissions apply only over the network; NTFS applies locally and remotely; the effective permission is the most restrictive of the two; an explicit deny wins
NTFS rules	Copying to another volume takes the destination's permissions; moving within a volume keeps the original; inheritance flows from the parent unless blocked; attributes: read-only, hidden, system, archive
UAC	User Account Control prompts before elevation; run as administrator only when needed; standard users enter admin credentials
BitLocker and To Go	Full-volume encryption keyed to the TPM (Pro and above); BitLocker To Go for removable drives; save the recovery key
EFS	Encrypting File System: per-file or per-folder encryption tied to the user's certificate; lose the key, lose the files

Active Directory tasks

- Join the domain from System settings; the computer object appears in **AD**; users then log in with domain accounts
- Login scripts run at sign-in to map drives and printers; home folders give each user a server folder; folder redirection points Documents and Desktop to the server for backup
- Organizational units group users and computers so Group Policy can be applied per **OU**; move objects between **OU**s to change what applies
- Security groups hold users for permissions; assign permissions to groups, never to individuals; Group Policy enforces settings across the domain

Workstation hardening

- Data-at-rest encryption (BitLocker, FileVault); **BIOS** and **UEFI** passwords
- Passwords: length first, then character types, uniqueness, complexity, expiration; password managers instead of reuse
- Users: screensaver locks, log off when away, secure laptops with cable locks, protect **PII** and passwords
- Accounts: restrict permissions, restrict login times, disable the guest account, failed-attempt lockout, timeout and screen lock, account expiration dates
- Change the default administrator account and password; disable AutoRun and AutoPlay; disable unused services

Mobile device security

- Device encryption; screen locks (facial recognition, **PIN**, fingerprint, pattern, swipe; swipe is not security); configuration profiles pushed by **MDM**
- Patch management for the **OS** and apps; endpoint security (antivirus, antimalware, content filtering)
- Locator applications, remote wipe, remote backup, failed-login restrictions (wipe after N attempts)
- Policies: **MDM** enrollment, **BYOD** versus corporate-owned, profile security requirements

10 SOHO, browser, disposal, and the malware removal procedure

SOHO router security

- Change default passwords; update firmware; secure management access (**HTTPS**, disable remote or **WAN** management); physical placement in a secure spot
- **IP** filtering and **MAC** filtering; content filtering; disable UPnP; screened subnet (**DMZ**) for exposed servers
- Wireless: change the **SSID**, disabling **SSID** broadcast hides but does not secure, **WPA3** or **WPA2-AES**, guest network isolated from the **LAN**
- Firewall: disable unused ports, port forwarding or mapping only for needed services

Browser security

- Download from trusted sources; verify hashes; avoid untrusted sources; keep the browser patched
- Extensions and plug-ins only from trusted stores; review permissions; remove unknown ones
- Password managers; valid certificates (padlock, no warnings); pop-up blocker; clear browsing data and cache; private browsing mode; sign-in and sync across devices; ad blockers; proxy; secure **DNS** (**DNS over HTTPS**)
- Browser feature management: enable or disable plug-ins, extensions, and features by policy

DISPOSAL METHOD	FACTS
Drilling, shredding, incineration	Physical destruction; certain; nothing recoverable
Degaussing	A strong magnet erases magnetic drives and tape; useless on SSDs
Erasing and wiping	Overwriting or secure erase; keeps the drive usable for reuse
Low-level format	Factory-level reset of the drive surface; modern drives do it in firmware
Standard format	Rebuilds the filesystem; a quick format leaves data recoverable
Outsourcing	Third-party vendor with a certificate of destruction or recycling

DISPOSAL METHOD	FACTS
Regulations	E-waste rules and data laws decide what must be destroyed and how

Malware removal, in order

- 1 Investigate and verify malware symptoms. 2 Quarantine the infected system (disconnect it from the network). 3 Disable System Restore in Windows Home. 4 Remediate: update the antimalware software, scan and remove (safe mode, preinstallation environment), reimagine or reinstall if needed. 5 Schedule scans and run updates. 6 Enable System Restore and create a restore point in Windows Home. 7 Educate the end user.

11 Software troubleshooting

WINDOWS SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Blue screen of death	Driver, hardware, or memory fault; note the stop code; recent changes; safe mode; roll back the driver; memory test
Degraded performance	Too many startup items, low RAM , full or failing disk, malware, background updates; Task Manager and Resource Monitor
Boot issues	Corrupted boot files or wrong boot order; Startup Repair in WinRE; bootrec; check the drive
Frequent shutdowns	Overheating, failing power supply, driver; check temperatures and Event Viewer
Services not starting	Dependency or account problem; Services console, Event Viewer, sfc
Applications crashing	Update or reinstall the app; check Event Viewer; compatibility mode; sfc
Low memory warnings	Not enough RAM or paging file; close apps; add RAM ; check for leaks
USB controller resource warnings	Too many devices on one controller; move devices to another port or a powered hub
System instability	Bad drivers, failing hardware, corrupt system files; sfc, DISM , memory test, roll back updates
No OS found	Boot order, unplugged drive, corrupted boot sector, USB stick left in; firmware settings; Startup Repair
Slow profile load	Large profile, roaming profile, login scripts, mapped drives timing out, corrupt profile (rebuild it)
Time drift	CMOS battery or a time source failure; sync with the domain or NTP ; certificates and Kerberos fail when time is off

Windows repair toolbox

- Safe mode (minimal drivers), WinRE (Startup Repair, System Restore, Command Prompt, Reset this PC), bootrec /fixmbr /fixboot /rebuildbcd, sfc /scannow, **DISM**, chkdsk /r, memory diagnostic, driver rollback, uninstall updates, rebuild the user profile, reinstall or repair install

MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
App fails to launch, close, or crashes	Force stop, clear cache, update, reinstall, restart the device
App fails to update or install	Storage full, OS version too old, no network, store account problem, MDM restriction
Slow to respond	Storage full, too many background apps, old OS , overheating, malware; restart
OS fails to update	Storage, battery level, network, unsupported (EOL) device
Battery life issues	Screen brightness, background apps, radios, poor signal, aged battery, malware
Random reboots	Bad app, OS bug, overheating, failing battery; update, remove recent apps, factory reset last

MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Connectivity issues	Bluetooth: pair again; Wi-Fi: forget and rejoin, airplane mode; NFC: enable it, remove the case
Screen does not autorotate	Rotation lock is on; then the sensor or a calibration

SECURITY SYMPTOM	MEANING
Mobile: unofficial app stores, developer mode, root or jailbreak	Bypass the platform's protections; malware and unsupported by MDM
Mobile: unauthorized or malicious app, application spoofing	A fake copy of a real app; remove it, change passwords
Mobile: high network traffic, degraded response, data-usage warnings, ads, fake security warnings, unexpected behavior, leaked files	Malware or a malicious app; remove, scan, reset if needed
PC: cannot access the network, desktop alerts, false antivirus alerts, altered or missing or renamed files, cannot access files, unwanted notifications, update failures	Infection or a rogue security program; run the malware removal procedure
Browser: random pop-ups, certificate warnings, redirection, degraded performance	Adware, a hijacked browser, a bad extension, or an on-path attack; reset the browser, remove extensions, check DNS and hosts file

12 Documentation, change management, and backups

Documentation and support

- Ticket fields: user information, device information, description of the issue, category, severity, escalation level; write clearly: issue description, progress notes, resolution
- Asset management: inventory lists, configuration management database (CMDB), asset tags and IDs, procurement life cycle, warranty and licensing, assigned users
- Documents: incident reports, standard operating procedures (SOPs), custom software installation procedures, new-user onboarding checklist, user off-boarding checklist, service-level agreements (internal and external or third-party), knowledge base articles

Change management

- Documented business processes: rollback plan, backup plan, sandbox testing, responsible staff members
- Request form: purpose, scope, change type (standard is pre-approved and routine, normal goes to the board, emergency is fixed first and reviewed after), date and time, change freeze periods, maintenance windows
- Affected systems and impact, risk analysis and risk level, change board approvals, implementation, peer review, end-user acceptance

BACKUP TERM	MEANING
Full	Everything; slowest to make, fastest to restore
Incremental	Only what changed since the last backup of any kind; small and fast; restore needs the full plus every incremental in order
Differential	Everything changed since the last full; grows daily; restore needs the full plus the latest differential
Synthetic full	A full backup assembled on the backup server from the last full and later incrementals, without touching the client
Recovery	In-place or overwrite restores over the original; alternative location restores to a new folder or machine to compare
Testing	Restore regularly; a backup never tested is not a backup; set a frequency
Rotation	On-site (fast) versus off-site (disaster); grandfather-father-son: daily, weekly, monthly sets; 3-2-1 rule: three copies, two media types, one off-site

13 Safety, environment, policy, and professionalism

Safety

- **ESD**: wrist strap clipped to the chassis, **ESD** mat, antistatic bags, touch bare metal first; humidity around 40 to 60 percent reduces static; never wear a strap inside a power supply or **CRT**
- Electrical: equipment grounding; disconnect power before repairing a **PC**; do not open power supplies or **CRTs** (capacitors)
- Handling and storage: cable management, antistatic bags, hold boards by the edges; comply with government regulations
- Personal: lifting with the legs and asking for help on heavy items; fire safety (class C extinguisher for electrical; pull the plug first); safety goggles; air filter mask for dust and toner

Environment

- Safety data sheet (**MSDS** or **SDS**) for handling and disposal of chemicals, toner, batteries
- Disposal: batteries to a recycler (lithium is a fire hazard), toner cartridges returned or recycled, monitors and devices to e-waste; never the trash
- Temperature and humidity within the vendor's range; ventilation; placement away from heat, dust, and moisture; dust cleanup with compressed air outdoors or a vacuum rated for electronics or toner
- Power: surges, brownouts (sags), blackouts; a **UPS** bridges outages and conditions power; a surge suppressor clamps spikes (joule rating)

Prohibited content, incident response, licensing, and privacy

- Incident response: preserve the scene, document, chain of custody (who held the evidence, when), inform management and law enforcement as required, make a forensic copy of the drive (hash it for integrity), collect in order of volatility (**CPU** registers and cache, memory, swap and temporary files, disk, remote logs, archives)
- Licensing: valid licenses only; **DRM** enforces them; **EULA** is the contract; perpetual (buy once) versus subscription; personal-use versus corporate-use; open-source licenses allow use and modification under their terms
- **NDA** and mutual **NDA** protect confidential information both ways
- Regulated data: credit card payment information (**PCI DSS**), personal government-issued information, **PII**, healthcare data (**HIPAA**); data retention requirements set how long records are kept
- Acceptable use policy; regulatory and business compliance; splash screens (login banners) state the rules before access

Communication and professionalism

- Professional appearance matching the environment (formal or business casual); proper language without jargon, acronyms, or slang; positive attitude and confidence
- Actively listen, do not interrupt; be culturally sensitive; use professional titles; be on time, or call if late; avoid distractions (personal calls, texting, social media, interruptions)
- Difficult situations: do not argue or be defensive, do not dismiss or judge, clarify with open-ended questions and restate the issue, use discretion about what you discuss
- Set and meet expectations, communicate status, offer repair or replacement options, document the service, follow up later; handle confidential material on screens, desks, and printers with discretion

14 Scripting, remote access, and AI

SCRIPT TYPE	RUNS IN
.bat	Windows batch, cmd.exe; oldest and simplest
.ps1	PowerShell; object-based, the Windows administrator's tool; execution policy may block scripts
.vbs	VBScript; legacy Windows scripting (deprecated)
.sh	Shell (bash) on Linux and macOS
.js	JavaScript; browsers and Node

SCRIPT TYPE	RUNS IN
.py	Python; cross-platform automation
Use cases	Basic automation, restarting machines, remapping network drives, installing applications, automated backups, gathering information, initiating updates
Considerations	Scripts can unintentionally introduce malware, inadvertently change system settings, or crash a browser or system by mishandling resources; test in a sandbox, read before running
REMOTE ACCESS	FACTS
RDP	Remote Desktop Protocol, TCP 3389 ; full Windows desktop; host needs Pro or above; use with a VPN or gateway, never open to the internet
VPN	Encrypted tunnel to the office network; then use any tool inside
VNC	Cross-platform screen sharing, port 5900; weak security unless tunneled
SSH	Secure shell, TCP 22 ; encrypted command line for Linux, macOS , network gear
RMM	Remote monitoring and management agents used by service providers; patching, alerts, remote control
SPICE	Simple Protocol for Independent Computing Environments: remote display for virtual machines (KVM and QEMU)
WinRM	Windows Remote Management: PowerShell remoting over HTTP 5985 and HTTPS 5986
Third-party tools	Screen-sharing, videoconferencing, file transfer, desktop management software; check their security and who can connect
Security	Encryption, MFA , least privilege, logging, closing ports when done, verifying the person on the other end

Artificial intelligence basics

- Application integration: **AI** assistants inside office, help desk, and code tools; policy sets appropriate use and forbids plagiarism (passing **AI** output off as one's own)
- Limitations: bias from training data, hallucinations (confident fabrications), accuracy that must be verified
- Private versus public **AI**: public services may store and learn from what you type; never paste confidential data; data security, data source, and data privacy decide which tool is allowed

15 Acronyms

Every acronym used in this course, 194 in all. In lessons and on this sheet, hover over an acronym for the short version; click it, or click one below, for the longer explanation.

ACRONYM	STANDS FOR	MEANING
5G	Fifth-generation cellular	The current mobile network generation with higher speed and lower latency than LTE.
802.1X	IEEE 802.1X	Port-based network access control: a device must authenticate before the switch port or Wi-Fi passes traffic.
ABC	Class A, B, and C fire extinguisher	A multipurpose extinguisher for ordinary combustibles, flammable liquids, and electrical fires.
AC	Alternating current	Mains power, which reverses direction 60 times a second in North America.
ACL	Access control list	An ordered list of permit and deny rules on a router, firewall, or file system.
AD	Active Directory	Microsoft's directory service: the domain's database of users, computers, and groups.
AES	Advanced Encryption Standard	The standard symmetric cipher, with 128, 192, or 256-bit keys.
AES-CCMP	AES in CCMP mode	The AES-based encryption WPA2 uses for Wi-Fi.
AI	Artificial intelligence	Software that learns patterns and generates text, images, or decisions.
AP	Access point	The radio that bridges Wi-Fi clients onto the wired network.
APFS	Apple File System	The modern macOS and iOS file system with snapshots and encryption.
APN	Access point name	The gateway setting a phone uses to reach a carrier's data network.

ACRONYM	STANDS FOR	MEANING
ARP	Address Resolution Protocol	Finds the MAC address that goes with a known IPv4 address on the local segment.
AUP	Acceptable use policy	The rules for how staff may use company systems and data.
BEC	Business email compromise	Fraud by email impersonating an executive or vendor to redirect payments.
BIOS	Basic Input/Output System	The original PC firmware: 16-bit, text setup, MBR disks up to 2 TB.
BSOD	Blue screen of death	The Windows stop error screen from a driver or hardware fault.
BYOD	Bring your own device	Staff use personal phones and laptops for work.
CAD	Computer-aided design	Engineering and drafting software that needs a workstation-class GPU and RAM.
CCMP	Counter Mode with Cipher Block Chaining Message Authentication Code Protocol	The AES-based encryption WPA2 uses.
CEO	Chief executive officer	The top executive; the person impersonated in whaling and business email compromise.
CFO	Chief financial officer	The executive in charge of money; a common target and impersonation in payment fraud.
CMDB	Configuration management database	The inventory of IT assets and how they relate, used in change and incident management.
CMOS	Complementary metal-oxide semiconductor	The small battery-backed memory that keeps firmware settings and the clock.
CO2	Carbon dioxide	The gas in CO2 fire extinguishers, safe on electrical fires; in medicine, the gas measured by capnography.
CPU	Central processing unit	The processor that executes instructions.
CRT	Cathode ray tube	The old glass-tube monitor; holds dangerous voltage even when unplugged.
DC	Domain controller	The server that runs Active Directory and authenticates domain logons.
DDoS	Distributed denial of service	Many sources, usually a botnet, flood a target at once.
DHCP	Dynamic Host Configuration Protocol	Hands out IP addresses, masks, gateways, and DNS servers automatically (UDP 67 and 68).
DISM	Deployment Image Servicing and Management	The Windows tool that repairs the component store and services images.
DLP	Data loss prevention	Tools that detect and block sensitive data leaving by email, web, USB, or cloud.
DMA	Direct memory access	Devices move data to and from memory without going through the CPU.
DMZ	Demilitarized zone	A screened subnet between the internet and the internal network for public-facing servers.
DNS	Domain Name System	Translates names such as www.example.com into IP addresses (UDP 53, TCP 53 for zone transfers).
DoH	DNS over HTTPS	DNS queries carried inside HTTPS on TCP 443 so they cannot be read or altered in transit.
DoS	Denial of service	One source floods or crashes a target so it cannot serve users.
DRM	Digital rights management	Technology that restricts copying and use of licensed content and software.
DSS	Data Security Standard	The PCI DSS rules for protecting payment card data.
DVD	Digital versatile disc	Optical media holding 4.7 GB, or 8.5 GB dual layer.
DWORD	Double word	A 32-bit registry value type.
EAP	Extensible Authentication Protocol	The framework that carries authentication methods inside 802.1X.
EDR	Endpoint detection and response	Endpoint agents that record activity, detect attacks, and let responders isolate and clean the machine.
EFS	Encrypting File System	Per-file and per-folder encryption built into NTFS, tied to the user's account.
EOL	End of life	The vendor stops selling the product.
ESD	Electrostatic discharge	Static electricity that can destroy chips; prevented with a wrist strap, mat, and antistatic bags.
eSIM	Embedded SIM	A SIM built into the phone and provisioned by software.
EULA	End-user license agreement	The license terms a user accepts to install software.
exFAT	Extended File Allocation Table	A FAT variant for flash media with no 4 GB file limit.
ext4	Fourth extended file system	The common Linux file system.
FAT	File Allocation Table	The family of simple file systems (FAT16, FAT32, exFAT).
FAT32	File Allocation Table 32-bit	The simple, universal file system for USB drives; 4 GB file limit, no permissions.
FRT	Facial recognition technology	Biometric authentication by face.
FTP	File Transfer Protocol	Plaintext file transfer on TCP 21 (control) and 20 (data); replace with SFTP or FTPS.
GB	Gigabyte	About a billion bytes; the unit for RAM, storage, and data caps.
GFS	Grandfather-father-son	A backup rotation with daily, weekly, and monthly sets kept for increasing periods.
GHz	Gigahertz	Billions of cycles per second; CPU clock speed and Wi-Fi band frequency.
GPO	Group Policy Object	A set of Windows settings applied to users and computers, locally or through Active Directory.
GPT	GUID Partition Table	The modern disk partition scheme: any disk size, up to 128 partitions, required for UEFI boot.
GPU	Graphics processing unit	The processor that renders images; on a card or built into the CPU.

ACRONYM	STANDS FOR	MEANING
GRUB	Grand Unified Bootloader	The Linux boot loader that presents the kernel menu.
GUI	Graphical user interface	Windows, icons, and a mouse, as opposed to a command line.
GUID	Globally unique identifier	A 128-bit random identifier used for partitions, objects, and records.
HDD	Hard disk drive	Spinning magnetic storage: cheap per terabyte, slower, and fragile.
HFS+	Hierarchical File System Plus	The older macOS file system, replaced by APFS.
HIPAA	Health Insurance Portability and Accountability Act	The US law protecting health information and setting privacy and security rules.
HKCU	HKEY_CURRENT_USER	The registry hive holding the logged-in user's settings.
HKLM	HKEY_LOCAL_MACHINE	The registry hive holding machine-wide settings.
HR	Human resources	The department that hires and separates staff; its system should drive account provisioning and removal.
HTTP	Hypertext Transfer Protocol	Web traffic in the clear on TCP 80.
HTTPS	Hypertext Transfer Protocol Secure	HTTP inside TLS on TCP 443.
IAM	Identity and access management	The systems and processes for creating accounts, authenticating users, and granting permissions.
ID	Identifier or identification	A name or number that uniquely labels a user, device, session, or record.
iOS	iPhone operating system	Apple's mobile operating system for iPhone.
IP	Internet Protocol	The Layer 3 protocol that addresses and routes packets between networks.
iPadOS	iPad operating system	Apple's tablet operating system, a variant of iOS.
IPsec	Internet Protocol Security	The suite that encrypts and authenticates IP packets for VPNs.
IPv6	Internet Protocol version 6	128-bit addresses in eight hex groups; no broadcast, built-in autoconfiguration.
IRQ	Interrupt request	The signal a device uses to get the CPU's attention.
IRQL	Interrupt request level	The priority level of Windows kernel code; a common stop code names it.
ISO	International Organization for Standardization; also an ISO disc image	The standards body, and the .iso file format that holds a complete disc image.
ISP	Internet service provider	The company that connects you to the internet.
IT	Information technology	Computers, networks, and the people who run them.
JIT	Just-in-time access	Privileges are granted only when needed and removed afterward.
KB	Kilobyte	About a thousand bytes.
KDC	Key Distribution Center	The Kerberos service on a domain controller that issues tickets.
KVM	Kernel-based Virtual Machine, or a keyboard-video-mouse switch	The Linux kernel's built-in hypervisor; also the switch that shares one console among computers.
L2TP	Layer 2 Tunneling Protocol	A VPN tunneling protocol that needs IPsec for encryption (L2TP/IPsec).
LAN	Local area network	One site's network: a building, floor, or campus that one organization owns and runs.
LDAP	Lightweight Directory Access Protocol	Queries a directory such as Active Directory for users and groups; TCP 389, secure 636.
LTO	Linear Tape-Open	The standard tape format for backups and archives.
MAC	Media access control (address)	The 48-bit hardware address burned into a network interface, used at Layer 2.
macOS	Mac operating system	Apple's desktop operating system.
MBR	Master boot record	The legacy partition scheme: disks up to 2 TB, four primary partitions, BIOS boot.
MBR2GPT	MBR to GPT conversion tool	The Windows command that converts a disk from MBR to GPT without data loss.
MDM	Mobile device management	Software that enrolls phones and tablets and enforces policy: passcodes, encryption, apps, remote wipe.
MDR	Managed detection and response	An outsourced team that monitors your EDR or XDR and responds around the clock.
MDT	Microsoft Deployment Toolkit	Free Microsoft tooling for building and deploying Windows images.
MFA	Multi-factor authentication	Two or more different kinds of proof: something you know, have, are, or where you are.
MMC	Microsoft Management Console	The framework that hosts administrative snap-ins such as Device Manager and Event Viewer.
MNDA	Mutual non-disclosure agreement	An NDA that binds both parties equally.
MSDS	Material safety data sheet	The older name for the safety data sheet.
NAS	Network-attached storage	A file server appliance that shares folders over the regular network using SMB or NFS.
NAT	Network address translation	Rewrites private inside addresses to a public address at the router so many hosts can share it.
NDA	Non-disclosure agreement	A contract that keeps shared information confidential.
NFC	Near-field communication	Very short range radio (about 4 cm) at 13.56 MHz for tap-to-pay and badge readers.
NFO	System Information file	The .nfo file saved by msinfo32 with a full hardware and software inventory.
NIC	Network interface card	The adapter that connects a computer to the network, wired or wireless.

ACRONYM	STANDS FOR	MEANING
NLA	Network Level Authentication	Remote Desktop authenticates the user before showing a logon screen.
NTFS	New Technology File System	The standard Windows file system: permissions, encryption, compression, large files, journaling.
NTP	Network Time Protocol	Synchronizes clocks across the network (UDP 123); wrong time breaks Kerberos, certificates, and logs.
NVMe	Non-Volatile Memory Express	The fast SSD interface that talks to PCIe lanes directly, usually in an M.2 slot.
OS	Operating system	The software that manages hardware and runs applications: Windows, macOS, Linux, iOS, Android.
OSHA	Occupational Safety and Health Administration	The US agency that sets workplace safety rules.
OTP	One-time password	A code that is valid for a single login.
OU	Organizational unit	A container in Active Directory used to group users and computers so policies and delegation can target them.
PAC	Proxy auto-configuration file	A script that tells the browser which proxy to use for each URL.
PAM	Privileged access management	Controls and records the use of administrator accounts.
PC	Personal computer	A desktop or laptop for one user.
PCI	Peripheral Component Interconnect	The older expansion bus; PCIe (PCI Express) is its modern serial successor.
PID	Process identifier	The number the operating system gives each running process.
PII	Personally identifiable information	Data that identifies a person: name, address, Social Security number, date of birth.
PIN	Personal identification number	A short numeric secret, a something-you-know factor.
POST	Power-on self-test	The firmware's hardware check at power-up; failures show as beep codes or a POST card reading.
PSK	Pre-shared key	One shared passphrase for everyone on a Wi-Fi network (Personal mode).
PSU	Power supply unit	Converts wall AC to the DC voltages a computer needs: 12 V, 5 V, 3.3 V.
PUP	Potentially unwanted program	Software that rides along with a download: toolbars, adware, bundled utilities.
PXE	Preboot Execution Environment	Booting a computer from the network to install or image an operating system.
QEMU	Quick Emulator	The open-source machine emulator often paired with KVM.
QR	Quick response code	The square barcode a phone camera reads.
RADIUS	Remote Authentication Dial-In User Service	The AAA server behind Wi-Fi Enterprise, VPNs, and 802.1X; UDP 1812 and 1813.
RAID	Redundant array of independent disks	Combining disks for speed, capacity, or fault tolerance: 0 stripe, 1 mirror, 5 stripe with parity, 6 double parity, 10 mirrored stripes.
RAM	Random-access memory	The working memory that holds running programs and data; lost at power off.
RDP	Remote Desktop Protocol	Microsoft's remote graphical desktop on TCP 3389.
RFID	Radio-frequency identification	Tags read by radio for badges, inventory, and asset tracking.
RHEL	Red Hat Enterprise Linux	Red Hat's commercially supported Linux distribution.
RMM	Remote monitoring and management	Tools that managed service providers use to monitor, patch, and remotely control many computers.
S . M . A . R . T	Self-Monitoring, Analysis, and Reporting Technology	Drive self-diagnostics that warn of impending failure.
SAE	Simultaneous authentication of equals	The WPA3 Personal handshake that resists offline password guessing.
SAML	Security Assertion Markup Language	An XML standard for passing authentication between an identity provider and a web application (enterprise SSO).
SCP	Secure Copy Protocol	Simple file copying over SSH, TCP 22.
SD	Secure Digital card	The flash memory card used in cameras and devices; miniSD and microSD are the smaller sizes.
SDS	Safety data sheet	The document describing a chemical's hazards, handling, and first aid; formerly MSDS.
SFTP	SSH File Transfer Protocol	File transfer inside an SSH session, TCP 22.
SHA-256	Secure Hash Algorithm 256-bit	The standard hash today: a 256-bit fingerprint of any input.
SIM	Subscriber identity module	The card or embedded chip that identifies a phone to the cellular network.
SLA	Service level agreement	The contract stating measurable service commitments, such as 99.9% uptime.
SMB	Server Message Block	Windows file and printer sharing on TCP 445.
SMS	Short Message Service	Text messages; used for one-time codes and also a phishing channel (smishing).
SOHO	Small office, home office	A small network with a consumer-grade router doing NAT, DHCP, Wi-Fi, and firewall in one box.
SOP	Standard operating procedure	A documented, step-by-step way of doing a routine task.
SPICE	Simple Protocol for Independent Computing Environments	An open remote display protocol for virtual machines.
SQL	Structured Query Language	The language used to query databases; also the name of the servers on TCP 1433 (Microsoft) and 3306 (MySQL).

ACRONYM	STANDS FOR	MEANING
SSD	Solid-state drive	Flash storage with no moving parts: fast, silent, and shock resistant.
SSH	Secure Shell	Encrypted remote command-line access and file transfer on TCP 22; replaces Telnet and FTP.
SSID	Service set identifier	The Wi-Fi network name that access points broadcast and clients join.
SSO	Single sign-on	Log in once and reach many systems without signing in again.
SSTP	Secure Socket Tunneling Protocol	Microsoft's VPN protocol that carries PPP inside TLS on port 443.
TACACS+	Terminal Access Controller Access-Control System Plus	Cisco's AAA protocol for device administration on TCP 49; encrypts the whole payload and separates authentication from authorization.
TB	Terabyte	About a trillion bytes; the unit for large drives.
TCP	Transmission Control Protocol	Reliable, ordered, connection-based transport: three-way handshake, acknowledgments, retransmission.
TGT	Ticket-granting ticket	The Kerberos ticket a user receives at logon, used to request tickets for services.
TKIP	Temporal Key Integrity Protocol	WPA's per-packet keying scheme; deprecated.
TLS	Transport Layer Security	The encryption layer under HTTPS and most secure protocols; use 1.2 or 1.3.
TOTP	Time-based one-time password	A six-digit code from an authenticator app that changes every 30 seconds.
TPM	Trusted Platform Module	A security chip on the motherboard that stores keys and measurements; required for BitLocker and Windows 11.
TRIM	TRIM command	Tells an SSD which blocks are no longer in use so it can erase them ahead of time.
UAC	User Account Control	The Windows prompt that asks for consent or an administrator password before a privileged change.
UDP	User Datagram Protocol	Connectionless, fast transport with no delivery guarantee; used for voice, video, DNS, DHCP.
UEFI	Unified Extensible Firmware Interface	Modern firmware that replaces BIOS: GPT disks, Secure Boot, mouse-driven setup.
UID	User identifier	The number Linux uses to identify each user account.
UNC	Universal Naming Convention	The \\server\share path format for network resources.
UPS	Uninterruptible power supply	A battery unit that keeps equipment running through a power cut and conditions the power.
URL	Uniform resource locator	A web address such as https://www.example.com/page.
USB	Universal Serial Bus	The standard port for peripherals, storage, and charging.
VLAN	Virtual local area network	A logical LAN on a switch: ports in different VLANs are separate broadcast domains.
VM	Virtual machine	A complete computer running as software on a hypervisor.
VNC	Virtual Network Computing	A cross-platform remote desktop protocol.
VPN	Virtual private network	An encrypted tunnel across an untrusted network such as the internet.
VRAM	Video RAM	The memory on a graphics card that holds frames and textures.
WAN	Wide area network	Connects LANs across cities or countries, usually over links leased from a provider.
WEP	Wired Equivalent Privacy	The original Wi-Fi encryption, broken in minutes; never use it.
Wi-Fi	Wireless fidelity	The trade name for IEEE 802.11 wireless networking.
WLAN	Wireless local area network	A LAN delivered over Wi-Fi instead of cables.
WMI	Windows Management Instrumentation	The Windows framework for querying and managing system information, used by scripts and monitoring tools.
WPA	Wi-Fi Protected Access	The interim fix for WEP that used TKIP; deprecated.
WPA2	Wi-Fi Protected Access 2	Wi-Fi security using AES with CCMP; still common and acceptable.
WPA3	Wi-Fi Protected Access 3	The current Wi-Fi security standard: SAE in Personal mode, stronger encryption, protected management frames.
WPAD	Web Proxy Auto-Discovery	Lets a browser find its proxy settings automatically through DHCP or DNS.
WS-Management	Web Services for Management	The standard protocol behind Windows Remote Management and PowerShell remoting.
WWAN	Wireless wide area network	Cellular data connectivity, such as a laptop's built-in LTE or 5G modem.
XDR	Extended detection and response	EDR extended across endpoints, network, cloud, and email into one correlated view.
XFS	XFS file system	A high-performance journaling file system used by Red Hat Enterprise Linux and others.
XSS	Cross-site scripting	An attacker's script runs in other users' browsers because a site echoes untrusted input.